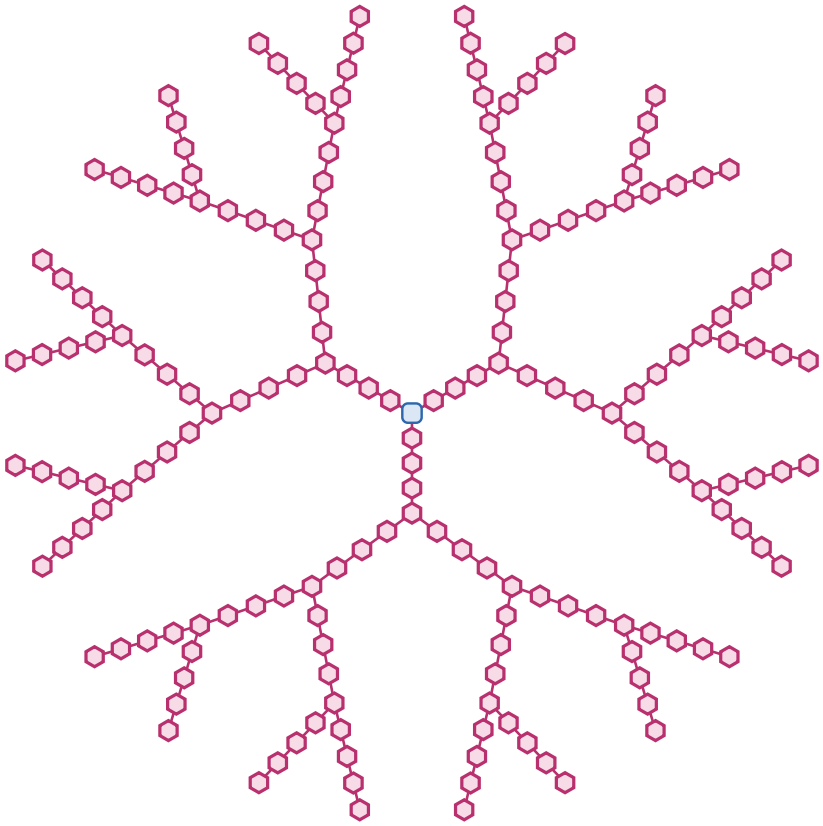


Glycogenesis & Glycogenolysis

How your body saves glucose, and how it spends it. Explained in plain words, with diagrams, real photos, real-life examples and animations.

glucose & glycogen · building (fed · insulin) · breaking down (fasting · glucagon) · phosphate

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How to use this pack

Read the **big idea** page first. Then work through the two pathways with the diagrams, and watch the animations on your phone (scan the code). Finish with the **test yourself** page and the one-page summary.

Watch the animations
Scan with your phone camera. The same animations are in this folder as `glycogen-animations.mp4` and three GIFs.

What your subject outline asks

Glycogen synthesis: glycogen synthase adds UDP-glucose to a primer (branches made by branching enzyme). Breakdown: glycogen phosphorylase removes glucose-1-phosphate (debranching enzyme handles branches). Reciprocal regulation by insulin vs glucagon/adrenaline via the cAMP/PKA cascade. Diseases: von Gierke and McArdle. Exam focus: *compare hormonal control of liver vs muscle glycogen.*

START HERE

The big idea in one page

Your brain and red blood cells need a steady supply of glucose, every minute, even while you sleep. But you only eat a few times a day. **Glycogen** is how your body solves this: it is a **glucose savings account**.

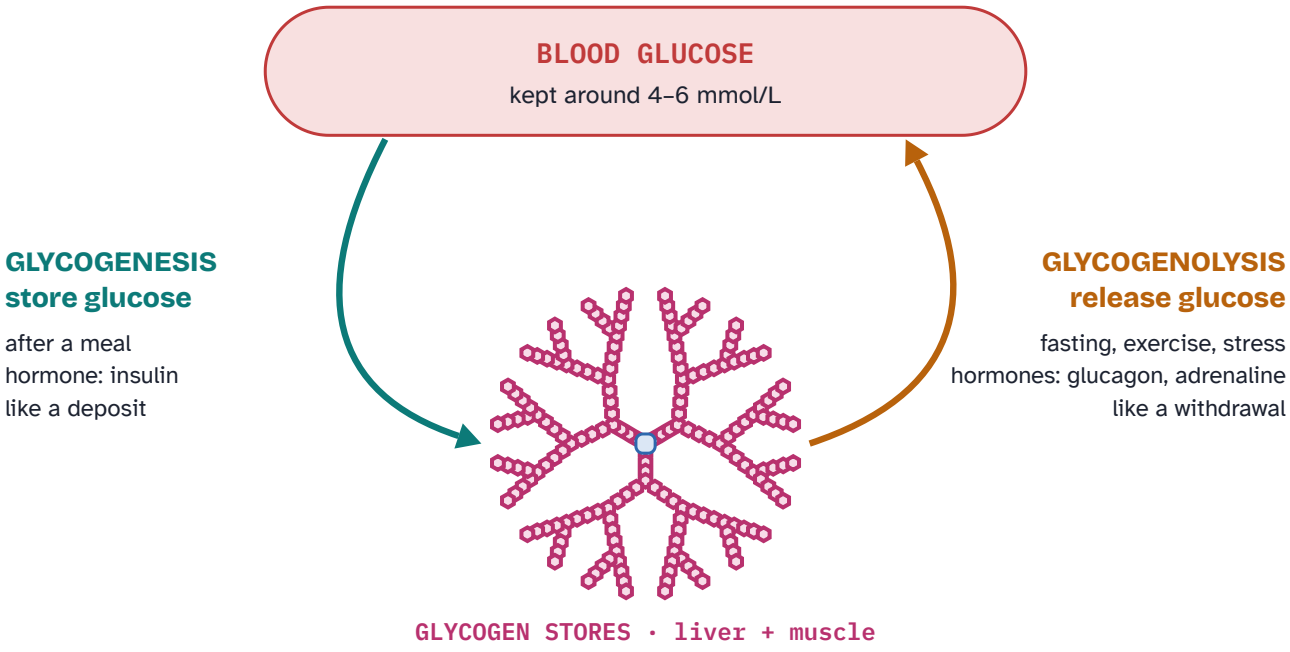


Fig. 1 · After a meal you deposit glucose into glycogen. Between meals you withdraw it.

glyco (sugar) **gen** (make) **esis** (creation)

Glycogenesis = making glycogen from glucose.

glyco (sugar) **gen** (make) **o** **lysis** (splitting)

Glycogenolysis = splitting glycogen back into glucose.

In plain words

Glycogen is a big, bushy tree made of thousands of glucose units joined together. Build the tree when there is extra glucose. Cut pieces off the tree when glucose runs low.

Don't mix these up

Glycogenolysis = breaking down glycogen.
Gluconeogenesis = making *brand new* glucose from lactate, glycerol or amino acids. Different pathway (also Week 4).

LOCATION

Where glycogen is stored

Almost every cell has a little glycogen, but two places hold nearly all of it: the **liver** and the **skeletal muscles**. They store it for very different reasons.

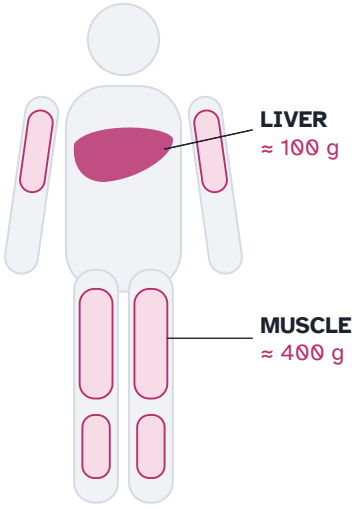
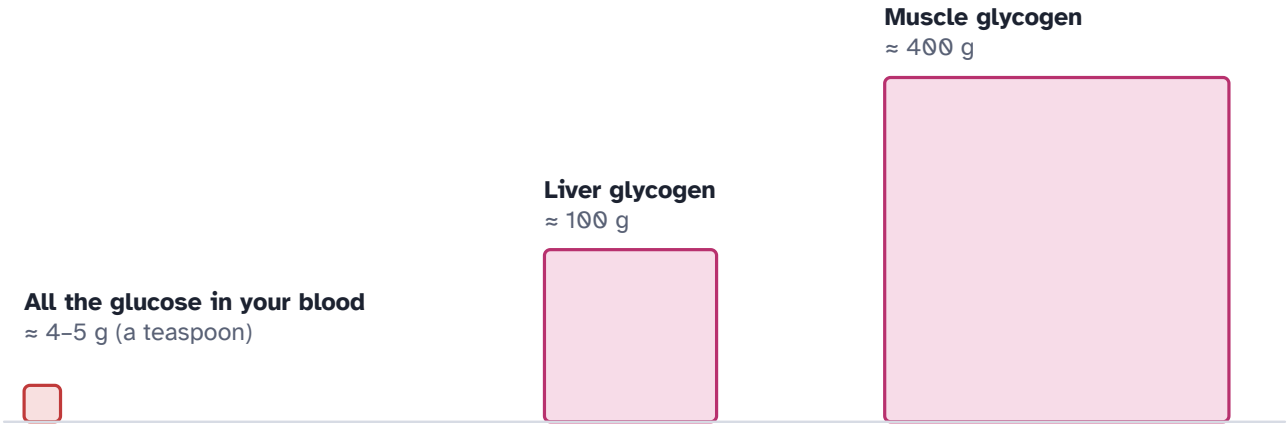


Fig. 2 · The two big glycogen stores.

	Liver	Muscle
Amount	≈ 100 g	≈ 400 g (all muscles)
Job	Keeps **blood** glucose steady for the whole body (brain!)	Fuel for the **muscle itself** when it contracts
Can release free glucose into blood?	**Yes**, has glucose-6-phosphatase	**No**, lacks glucose-6-phosphatase
Listens to glucagon?	Yes	No (no glucagon receptors)
Listens to adrenaline?	Yes	Yes
Lasts about	12–24 h of fasting	≈ 1.5–2 h of hard exercise



Drawn to scale: the area of each square is the grams of glucose it holds.

Fig. 3 · Your blood holds only a few grams of glucose at any moment. The glycogen stores are what keep it topped up.

Easy way to remember

The liver is generous: it shares its glucose with everyone. **Muscle is selfish:** it keeps its glycogen for its own work.

STRUCTURE

What glycogen looks like

Glycogen is a **branched polymer of glucose**. Think of a tree: long chains are the branches, and every chain splits again and again.

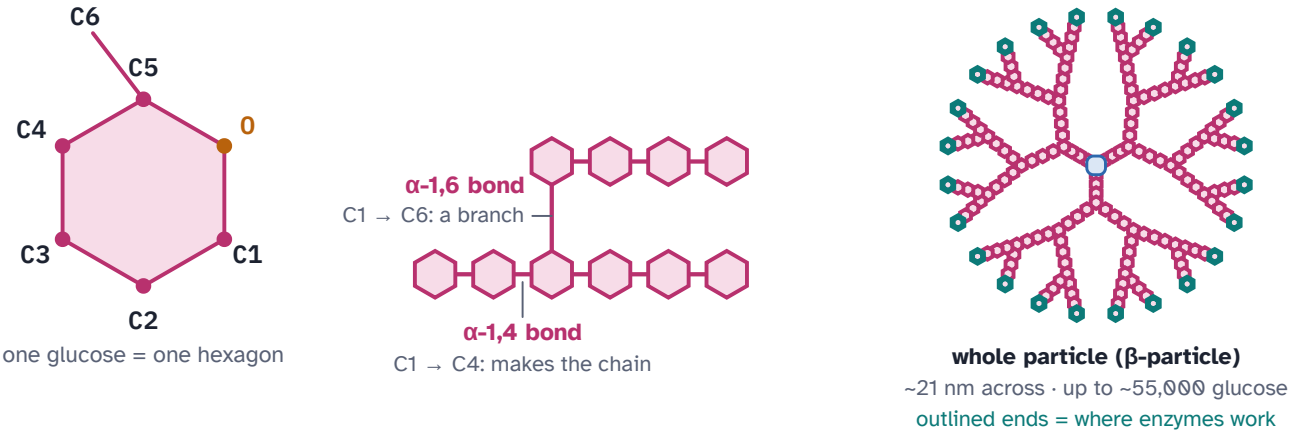
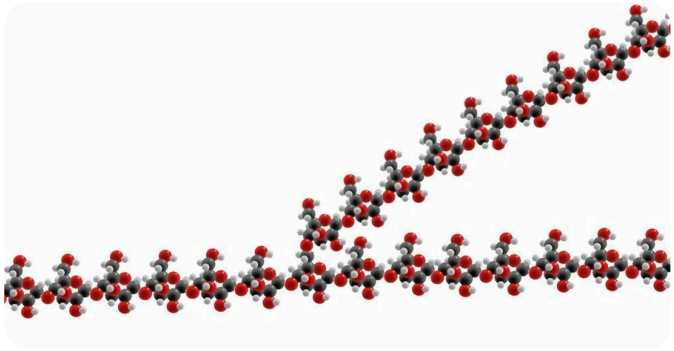


Fig. 4 · Left: carbons in one glucose. Middle: α-1,4 bonds make chains; α-1,6 bonds make branches. Right: a whole particle built around one glycogenin protein.

- **Chains** use **α-1,4** glycosidic bonds (carbon 1 of one glucose to carbon 4 of the next).
- **Branches** use **α-1,6** bonds, about every 8–12 glucose.
- A full particle can hold **up to ~55,000 glucose** and is about **21 nm** across. In the liver, particles clump into bigger “rosettes”.
- The ends of the outer chains are called **non-reducing ends**. All the building and cutting happens there.
- At the very centre sits **glycogenin**, the protein that started the particle.



A real computer model of one branched chain of glycogen, atom by atom (red = oxygen, black = carbon, white = hydrogen). You can see the side branch leaving the main chain.

CeresVesta, Wikimedia Commons, public domain

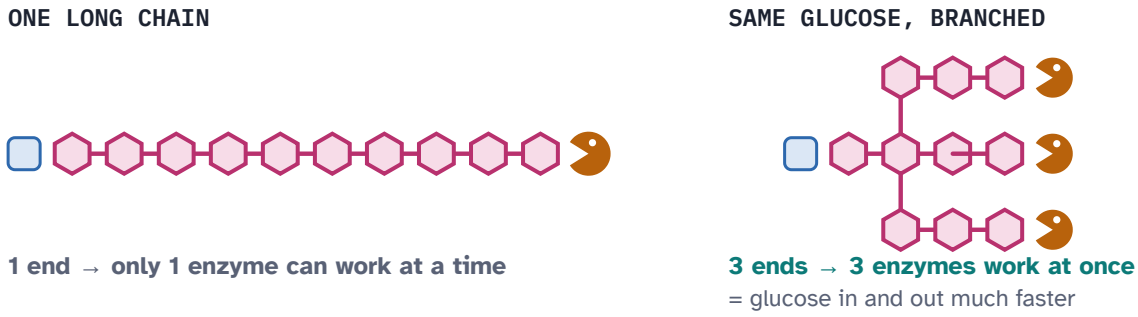


Fig. 5 · Why branch? More ends means more enzymes can work at the same time, so glucose goes in and out much faster. Branching also makes glycogen more soluble.

WHY BOTHER?

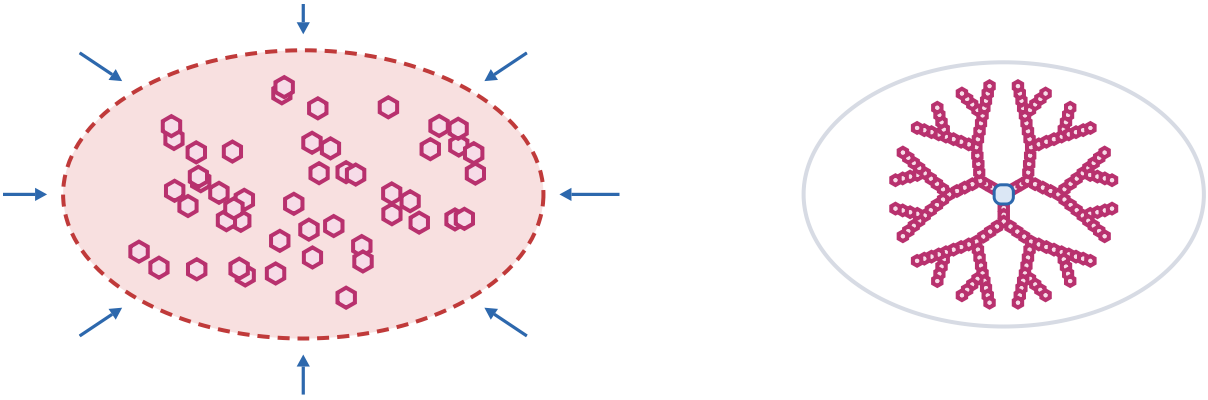
Why store glucose as glycogen?

Reason 1 · It keeps cells from swelling

Water moves toward where there are more dissolved particles (osmosis). The glucose stored in a liver cell would be about **0.4 mol/L** if it floated around free, and water would rush in until the cell swelled and burst. Packed into glycogen, the same glucose counts as only a few huge particles.

Reason 2 · It is fast

Fat stores far more energy, but it is slow to use and **can't be turned into glucose**. Glycogen can be broken down within seconds, and muscle can burn it even without oxygen (sprinting).



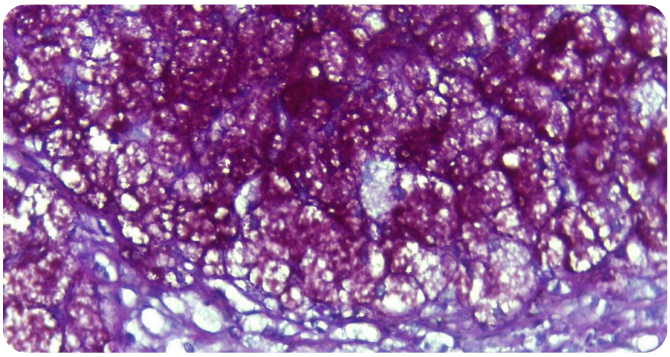
free glucose: water rushes in → cell swells

same glucose as ONE glycogen particle: cell is fine

Fig. 6 · Same amount of glucose, two ways of storing it.

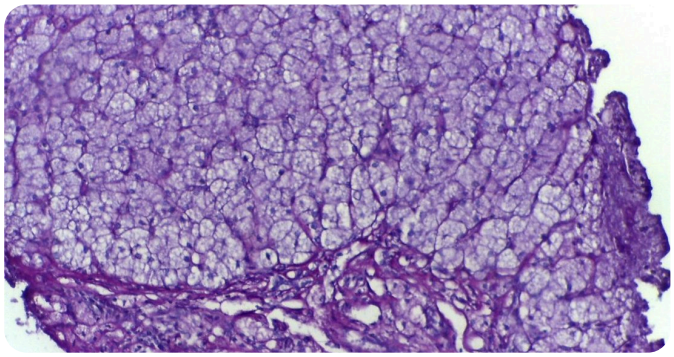
See glycogen with your own eyes: the PAS stain

Pathologists stain tissue with **PAS** (periodic acid–Schiff). It turns glycogen **magenta**, which is why glycogen is magenta in every diagram in this pack. To prove the pink stuff really is glycogen, they digest a second slice with **diastase** (an enzyme that breaks glycogen down) and stain again.



Liver biopsy, PAS stain. The liver cells are packed with glycogen (deep magenta). This patient has a glycogen storage disorder, so there is far more than normal.

Dept. of Pathology, Calicut Medical College, CC BY-SA 4.0



The same liver after diastase, then PAS. The magenta is gone: the enzyme digested the glycogen. This PAS / PAS-diastase pair is a real lab test.

Dept. of Pathology, Calicut Medical College, CC BY-SA 4.0

PATHWAY 1 · BUILD

Glycogenesis, step by step

When: **after a meal**, when blood glucose and **insulin** are high. Where: the **cytosol** of liver and muscle cells.



Glucose in the blood
enters through GLUT2 (liver) or GLUT4 (muscle; insulin moves GLUT4 to the surface)

1

Hexokinase (glucokinase in liver) `ATP → ADP`

Sticks a phosphate onto carbon 6. The negative charge traps glucose inside the cell.

Exam words: phosphorylation of glucose to glucose-6-phosphate (also step 1 of glycolysis).



Glucose-6-phosphate (G6P)

2

Phosphoglucomutase

Moves the phosphate from carbon 6 to carbon 1. Nothing is added or lost; it is just rearranged.

Exam words: isomerisation G6P ⇌ G1P; reversible.



Glucose-1-phosphate (G1P)

3

UDP-glucose pyrophosphorylase `UTP in` `PPi → 2 Pi`

Joins G1P to UTP to make **UDP-glucose**, the "activated" glucose that is ready to be added. The leftover pyrophosphate (PPi) is chopped into 2 phosphates, which makes the step one-way.

Exam words: G1P + UTP → UDP-glucose + PPi; PPi hydrolysed by pyrophosphatase.



UDP-glucose
the activated donor: think of UDP as the "ticket" that lets glucose join the chain

4

Glycogen synthase `UDP out`

Adds the glucose to the end of an existing chain with an **α-1,4** bond. This is the **slowest, controlled step** (rate-limiting). It needs a primer made by glycogenin.

Exam words: rate-limiting enzyme; regulated by phosphorylation (inactive when phosphorylated).



Glycogen, one glucose longer (n + 1)

5

Branching enzyme

Cuts a block of ~7 glucose from a long chain and re-attaches it lower down with an **α-1,6** bond.

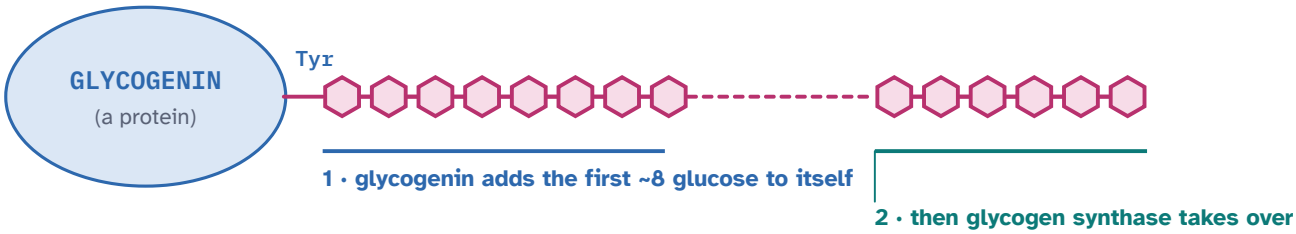
Exam words: amylo-(1,4→1,6)-transglycosylase.



Branched glycogen

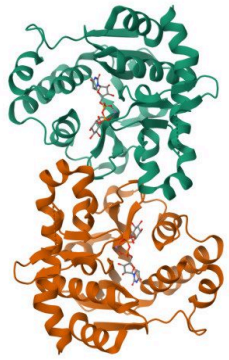
PATHWAY 1 · CLOSE-UPS

Glycogenesis close-ups



Glycogen synthase cannot start from nothing. It needs a short primer chain, and glycogenin makes that primer. Glycogenin stays in the centre of every particle.

Fig. 7 · Every glycogen particle starts on a protein called glycogenin.



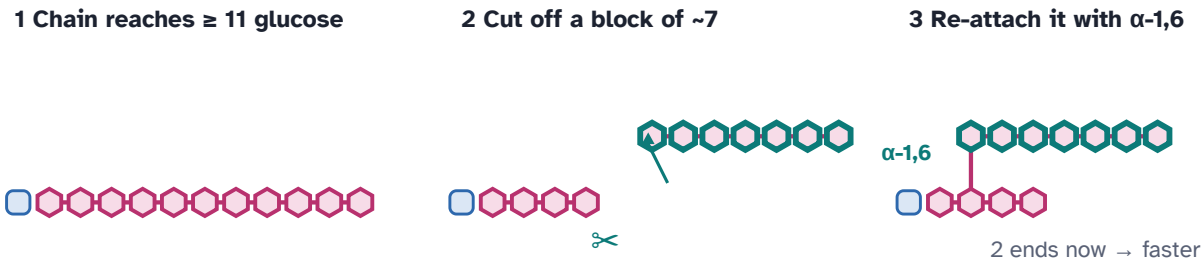
The real 3D structure of glycogenin (two copies, green and orange) holding UDP-glucose, solved by X-ray crystallography.
RCSB PDB, entry 1LL2 (Gibbons et al., 2002)

Glycogenin in plain words

Glycogen synthase can only *extend* a chain; it cannot start one. Glycogenin is the starter: it grabs UDP-glucose and attaches about 8 glucose to itself. Then synthase takes over.

Energy cost of storing glucose

From blood glucose to glycogen costs **2 high-energy phosphates per glucose**: 1 ATP (hexokinase) + 1 UTP (UDP-glucose). Getting it back out costs nothing, because phosphorylase uses free phosphate. Storing glucose as glycogen is therefore very efficient.



Rule of thumb: branch points end up about 8–12 glucose apart, and a new branch is at least 4 glucose away from the nearest one.

Fig. 8 · How branching enzyme builds the tree.

PATHWAY 2 · BREAK

Glycogenolysis, step by step

When: **between meals, overnight, during exercise or stress**. Triggered by **glucagon** (liver) and **adrenaline** (liver and muscle).

Glycogen (n glucose)

1

Glycogen phosphorylase

`Pi in` `helper: PLP (vitamin B6)`

Snips one glucose at a time off the chain ends. It uses **phosphate**, not water, so each glucose comes off already carrying a phosphate. It stops 4 glucose away from a branch.

Exam words: phosphorolysis of α-1,4 bonds → glucose-1-phosphate; rate-limiting step.

Glucose-1-phosphate + glycogen (n − 1)

2

Debranching enzyme

At each branch: moves 3 of the last 4 glucose to another chain end, then cuts the α-1,6 link to release the last one as **free glucose**.

Exam words: bifunctional: 4-α-glucanotransferase + α-1,6-glucosidase (hydrolysis).

3

Phosphoglucomutase

Moves the phosphate back from carbon 1 to carbon 6.

Exam words: same enzyme as step 2 of glycogenesis, running the other way.

Glucose-6-phosphate

If this is a LIVER cell

Step 4 · Glucose-6-phosphatase (in the endoplasmic reticulum) takes off the phosphate. Free glucose leaves through GLUT2 and **raises blood glucose**.

If this is a MUSCLE cell

There is **no glucose-6-phosphatase**. G6P stays trapped and goes straight into **glycolysis** → **ATP** for contraction (and lactate when oxygen is short).



Real 3D structure of glycogen phosphorylase, a pair of identical subunits (orange and green).
RCSB PDB, entry 1GPB

Why a pair?

The two halves talk to each other. When one half is switched on (by a phosphate or by AMP), it pulls the other into the active shape too. This is how the enzyme flips quickly between OFF and ON.

PATHWAY 2 · CLOSE-UPS

Glycogenolysis close-ups

1 Phosphorylase stops at 4 **2 Transferase moves 3** **3 Glucosidase frees the last 1**



Memory hook: 4 = 3 + 1. Of the 4 glucose left on the stub, 3 are moved and 1 is set free. About 1 in 10 glucose leaves glycogen this way.

Fig. 9 · Debranching enzyme does two jobs with one protein.

Why use phosphate instead of water?

Cutting with water (hydrolysis) would give plain glucose, and the cell would then have to spend ATP to put a phosphate back on. Cutting with phosphate (**phosphorolysis**) gives glucose-1-phosphate directly. That saves ATP, and the charged sugar cannot leak out of the cell.

Net ATP from glycolysis, per glucose

from blood glucose — 2 ATP
hexokinase spends 1 ATP first

from glycogen — 3 ATP
already has its phosphate: 1 ATP saved

Fig. 10 · Glucose from glycogen gives 1 more ATP in glycolysis.

The liver-versus-muscle fork is the key exam idea

	Liver	Muscle
Final product	Free glucose → blood	G6P → glycolysis → ATP
Why?	Has glucose-6-phosphatase	No glucose-6-phosphatase
Who benefits?	Brain, red blood cells, the whole body	Only that muscle
Main triggers	Glucagon (fasting), adrenaline (stress)	Adrenaline, Ca^{2+} and AMP from contraction

Common mistake

"Glycogenolysis produces glucose-6-phosphate, which then leaves the cell." **Wrong.** G6P can never leave a cell. Only free glucose can, and only the liver (and kidney) can make it.

Practice exam question (4 marks)

Q. In glycogen degradation, the last step is the change of glucose-1-phosphate to glucose-6-phosphate by phosphoglucomutase. Why is this step important?

Model answer. Glycogen phosphorylase releases glucose-1-phosphate, but the next pathways need glucose-6-phosphate. (1) In **muscle**, G6P enters glycolysis directly to make ATP for contraction. (2) In **liver**, G6P is the substrate for glucose-6-phosphatase, which releases free glucose to keep blood glucose up during fasting. Without this step, glycogen could not be used as fuel or to supply the blood.

COMPARE

Build vs break, side by side

Glycogenolysis is **not** glycogenesis run backwards. The two pathways share only phosphoglucomutase. The key steps use different enzymes, so each can be switched on or off by itself.

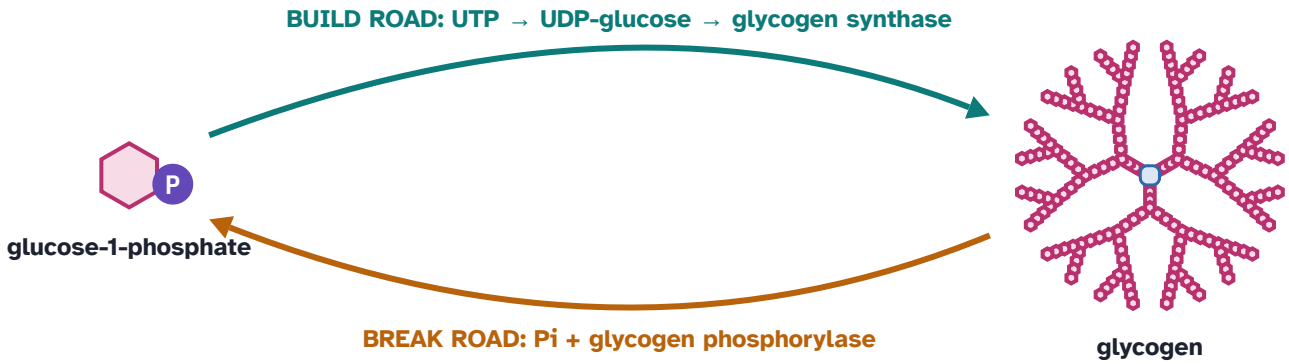


Fig. 11 · Two one-way roads. If both ran at full speed together, the cell would go round in circles and waste UTP (a “futile cycle”). Hormones make sure only one road is open.

	Glycogenesis (build)	Glycogenolysis (break)
When	Fed state, after meals	Fasting, exercise, stress
Hormone	Insulin	Glucagon (liver), adrenaline (liver + muscle)
Key enzyme	Glycogen synthase	Glycogen phosphorylase
Bond handled	Makes α-1,4 (synthase) and α-1,6 (branching enzyme)	Breaks α-1,4 (phosphorylase) and α-1,6 (debranching enzyme)
Glucose form used/made	Uses UDP-glucose	Makes glucose-1-phosphate (+ some free glucose)
Energy	Costs ATP + UTP	Uses free Pi, costs no ATP
Enzyme ON when	Synthase **dephosphorylated**	Phosphorylase **phosphorylated**
Helper molecule	Glycogenin primer	PLP (from vitamin B6)
Location	Cytosol	Cytosol (last step in liver: ER)

REGULATION 1

Who flips the switch? Hormones

Both key enzymes are controlled by **adding or removing a phosphate** (covalent modification). Hormones decide which one happens.

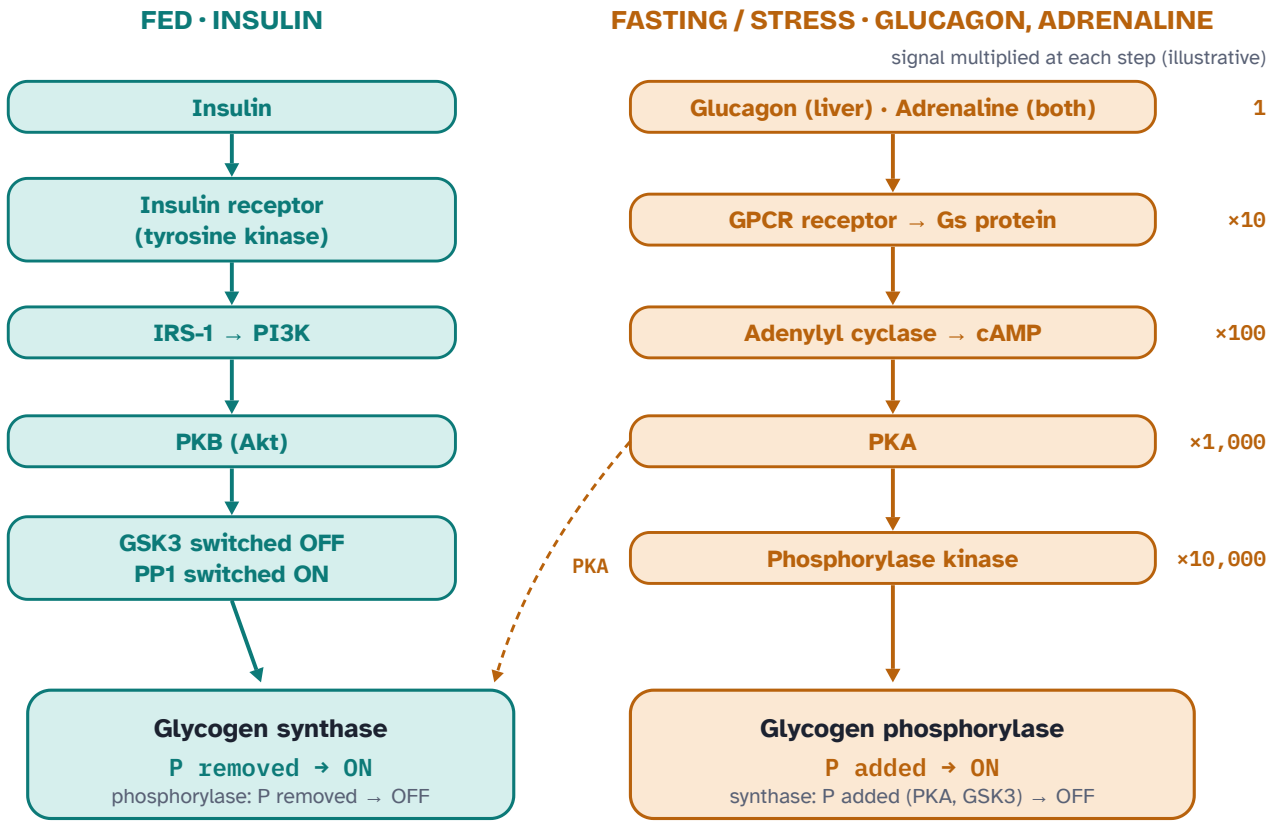


Fig. 12 · Left: insulin removes phosphates. Right: glucagon and adrenaline add them through the cAMP → PKA cascade.

The golden rule Phosphate **ON** → breakdown **ON**, building **OFF** (glucagon, adrenaline). Phosphate **OFF** → building **ON**, breakdown **OFF** (insulin).

Why a cascade?

Each enzyme in the chain switches on many copies of the next one. A handful of hormone molecules can therefore free huge amounts of glucose within seconds. That is exactly what you need when a fright hits.

Extra detail for top marks

PKA also switches on **inhibitor-1**, which blocks PP1, so the phosphates stay on. Insulin works the other way: PKB (Akt) inactivates GSK3 and PP1 becomes active, so synthase is dephosphorylated. Insulin also moves GLUT4 to the membrane in muscle and fat.

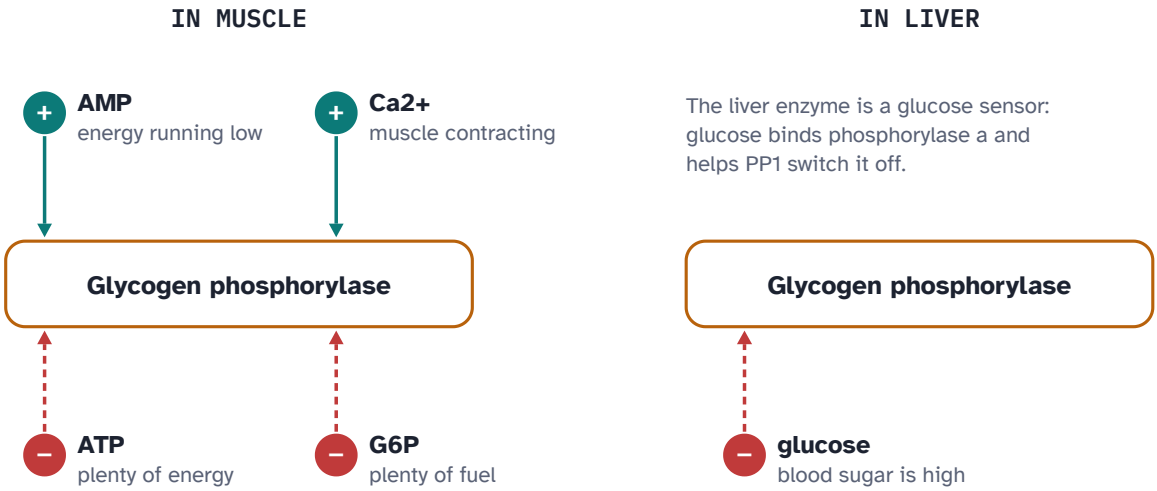
REGULATION 2

Fine control: sensors inside the cell

Besides hormones, small molecules inside the cell act as **allosteric** switches: they bind the enzyme and change its shape within milliseconds.

Enzyme	With phosphate	Without phosphate	Can be woken up by…
Glycogen synthase	“b” form · **OFF**	“a” form · **ON**	G6P can partly switch on the b form
Glycogen phosphorylase	“a” form · **ON**	“b” form · mostly **OFF**	AMP switches on the b form (muscle)

Tip: for both enzymes, **“a” = active**. But synthase *a* has **no** phosphate, while phosphorylase *a* **has** one.



In both tissues, G6P also switches ON glycogen synthase (lots of glucose inside → store it).

Fig. 13 · Green (+) switches the enzyme on; red (–) switches it off.

Muscle listens to its own work

When a nerve tells a muscle to contract, **Ca^{2+}** floods the cell. The same Ca^{2+} switches on phosphorylase kinase, so glycogen breakdown starts **at the same moment as contraction**, even before adrenaline arrives. Using ATP makes **AMP**, which pushes phosphorylase on even harder.

The liver tastes the blood

After a meal, glucose flows into the liver and binds phosphorylase *a*. This helps PP1 remove its phosphate and turn it off. The freed PP1 then switches glycogen synthase on. So high blood glucose flips the liver from “release” to “store”.

REAL LIFE 1

A normal day, and a fright



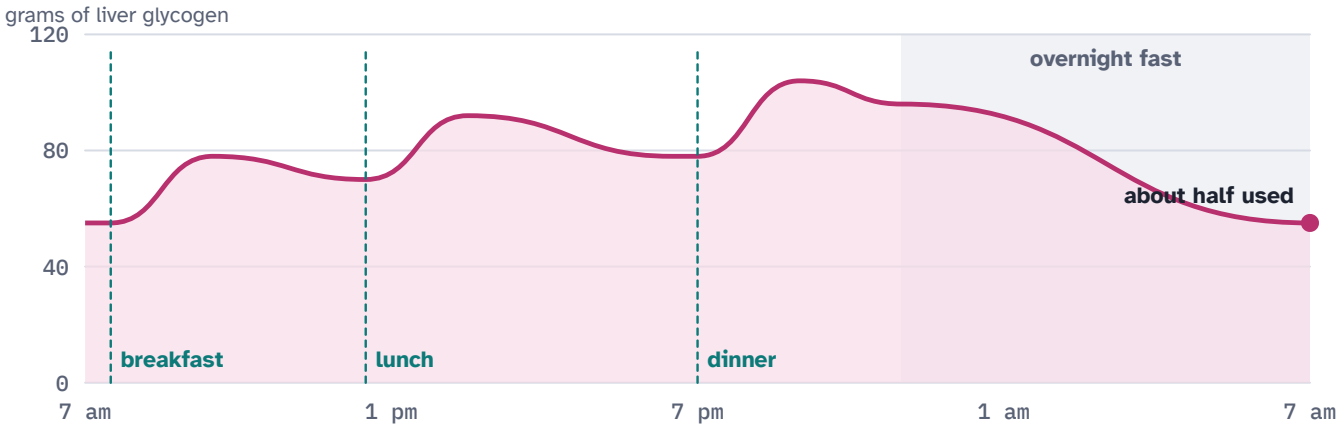
A plate of spaghetti has roughly 80–100 g of carbohydrate.
Wikimedia Commons, public domain

After lunch

The starch is digested to glucose, which enters the blood through the gut. The rise in blood glucose makes the pancreas release **insulin**. Liver and muscle take up glucose and **glycogen synthase** switches on: glycogenesis.

Overnight

No food for 10–12 hours, but your brain still uses about **120 g of glucose a day**. **Glucagon** rises, liver glycogen phosphorylase switches on, and the liver slowly feeds glucose into the blood. By breakfast, a large part of the liver's glycogen is gone. That is one reason breakfast feels important.



Schematic: the usual pattern described in textbooks, not one person's measured data.

Fig. 14 · Liver glycogen goes up after each meal and down overnight.

A fright: fight or flight

You step off the kerb and a car horn blasts. Within seconds your adrenal glands release **adrenaline**. In the liver it triggers the cAMP → PKA cascade and glucose pours into your blood. In your leg muscles it switches on glycogen phosphorylase so they have fuel to jump back. Your heart races, your hands shake, and your blood sugar rises, all from glycogenolysis.

REAL LIFE 2

Sprinting and marathons



Usain Bolt in a 100 m heat, 2013 World Championships in Moscow. A 100 m race lasts about 10 seconds.

Tobi 87, Wikimedia Commons, CC BY-SA 3.0

The 10-second sprint

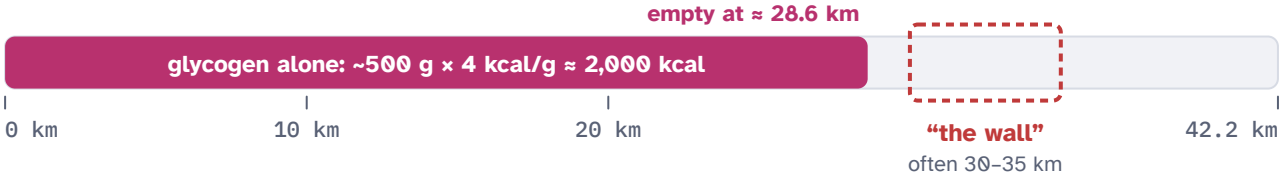
There is no time to wait for oxygen. Muscles burn stored ATP, phosphocreatine and **muscle glycogen**, turning it into lactate. The nerve signal releases **Ca^{2+}** and ATP use makes **AMP**, and both switch phosphorylase on instantly. Muscle glycogen gives 3 ATP per glucose unit in glycolysis instead of 2, which is a bonus when every bit of speed counts.

The 42.2 km marathon

This race is long enough to **run out of glycogen**. When muscle glycogen is nearly gone, the runner has to rely mostly on fat, which gives energy more slowly. The pace drops and the legs feel like lead. Runners call this “**hitting the wall**”.

Energy check for a 70 kg runner

Running costs about 1 kcal per kg per km → 70 × 42.2 ≈ 2,950 kcal for a marathon.



Fat supplies part of the energy, which pushes the wall later. Carb loading and sports gels top up glycogen.

Fig. 15 · Rough numbers show why glycogen alone cannot fuel a whole marathon.



Runners at the Berlin Marathon 2015. Several are carrying drinks; sports drinks and gels supply glucose during the race.

Denis Barthel, Wikimedia Commons, CC BY-SA 4.0

Carb loading, explained

For 1–2 days before a race, endurance athletes eat a lot of carbohydrate (sports guidelines suggest about 10–12 g per kg of body weight per day) and train lightly. With insulin high and the muscles resting, glycogen synthase fills the muscles well above their usual level. This is called “supercompensation”, and it can push the wall back or remove it.

REAL LIFE 3

Diabetes, diets and alcohol



Glucagon emergency kits. The powder is mixed with the liquid and injected when someone with diabetes has a severe hypo.

Mizi m, Wikimedia Commons, CC0

The glucagon rescue injection

A person with type 1 diabetes who injects too much insulin can have a dangerous **hypo** (low blood glucose, below about 4 mmol/L) and may become confused or unconscious. A friend can inject **glucagon** (or give a nasal spray). It works by triggering **liver glycogenolysis**, so blood glucose rises within about 10–15 minutes.

When it may NOT work

If the liver's glycogen is already used up, after a long fast, heavy drinking, or several hypos in a row, glucagon has nothing to release. This is a direct, real-world consequence of glycogenolysis.



Blood glucose meters. A normal reading is about 4–6 mmol/L; glycogen is a big reason it stays in that range between meals.

Christidy, Wikimedia Commons, public domain

Why low-carb diets cause fast early weight loss

Each gram of glycogen is stored with roughly **3 g of water**. In the first days of a very low-carb diet, glycogen is used up and the water leaves with it, so the scale can drop 1–2 kg quickly. It is mostly water, not fat.

Alcohol and low blood sugar

While the liver is busy breaking down alcohol, it makes new glucose (gluconeogenesis) poorly. If someone drinks heavily without eating, their liver glycogen runs out and blood glucose can fall dangerously low, even in people without diabetes.

CLINICAL LINK

When an enzyme is missing: glycogen storage diseases

A faulty enzyme blocks the pathway at one step. Most of these are inherited (autosomal recessive). Look at *where* the block is, and you can predict the symptoms.

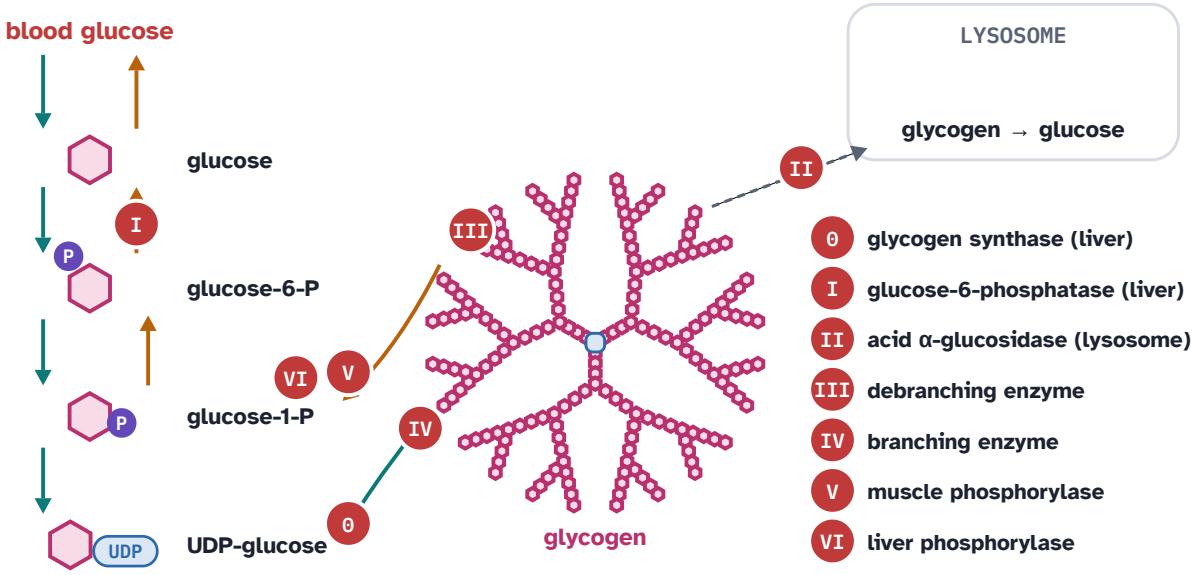
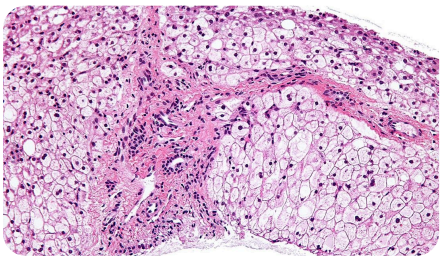


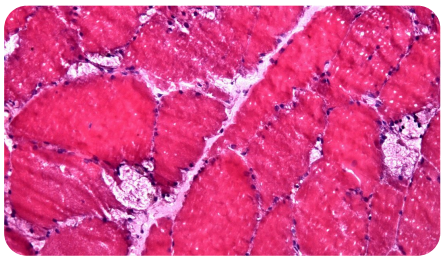
Fig. 16 · Each red badge shows which enzyme is missing in each type.

Type	Name	Missing enzyme	What you see, and why
I	**von Gierke**	Glucose-6-phosphatase (liver)	Severe **fasting hypoglycaemia** (the liver can't release glucose), big liver from stored glycogen, lactic acidosis. Treated with frequent feeds and uncooked cornstarch at night.
II	Pompe	Acid α-glucosidase (lysosome)	Glycogen builds up in lysosomes of heart and muscle: floppy baby, enlarged heart. Treated with enzyme replacement.
III	Cori	Debranching enzyme	Glycogen with short outer branches; big liver, milder hypoglycaemia.
IV	Andersen	Branching enzyme	Long, unbranched, poorly soluble glycogen damages the liver (cirrhosis).
V	**McArdle**	Glycogen phosphorylase (muscle)	**Exercise intolerance**, cramps, dark urine (myoglobin) after exertion; lactate does not rise with exercise. Blood glucose is normal because the liver is fine.
VI	Hers	Glycogen phosphorylase (liver)	Mild hypoglycaemia and a big liver.



Liver biopsy (H&E) in Cori disease: swollen liver cells with pale, glycogen-filled cytoplasm.

Nephron, Wikimedia Commons, CC BY-SA 3.0



Muscle biopsy in Pompe disease: the holes (vacuoles) are lysosomes stuffed with glycogen.

Jensflorian, Wikimedia Commons, CC BY-SA 3.0



Myozyme (alglucosidase alfa), an enzyme replacement treatment for Pompe disease.

Genzyme, Wikimedia Commons, public domain

REVISE

Memory tricks and exam traps

Synthase Sews, Phosphorylase Plucks
Synthase stitches glucose on. Phosphorylase plucks it off, using phosphate.

Insulin puts glucose in storage
Insulin = fed = glycogenesis.

Glucagon: the glucose is gone
Glucagon = fasting = liver glycogenolysis.

Phosphate = Panic mode
Phosphate on the enzymes (glucagon, adrenaline) = breakdown on, building off.

Liver is generous, muscle is selfish
Only the liver has glucose-6-phosphatase to release glucose.

4 = 3 + 1
Phosphorylase stops at 4. Debranching moves 3 and frees 1.

UDP is the ticket
Glucose must become UDP-glucose before synthase will add it.

Very Poor Carbohydrate Metabolism
Types I–III and V in order: von Gierke, Pompe, Cori, McArdle. Andersen (IV) sits between Cori and McArdle.

Exam traps

1. Glycogenolysis is **not** glycogenesis in reverse: different key enzymes, and UDP-glucose appears only in synthesis.
2. Phosphorylase uses **phosphate** (phosphorolysis). The only water-using cut is the α-1,6 step of debranching, which gives free glucose.
3. **Glucagon has no effect on muscle** (no receptors). Adrenaline acts on both.
4. “a” means active for both enzymes, but the phosphate is on phosphorylase *a* and **off** synthase *a*.
5. Muscle glycogen **cannot** raise blood glucose directly.
6. Glycogenolysis ≠ gluconeogenesis ≠ glycolysis. Say which one you mean!
7. Glycogenin is a **protein** primer. Synthase cannot start a chain on its own.
8. Glycogen uses α bonds; cellulose uses β-1,4 bonds, which humans cannot digest.

Mini glossary

Non-reducing end
the free outer end of a chain, where glucose is added or removed

Primer
a short starter chain that an enzyme extends

Covalent modification
switching an enzyme by adding or removing a group, here a phosphate

PLP
pyridoxal phosphate, the vitamin B6 helper inside glycogen phosphorylase

Phosphorolysis
cutting a bond with phosphate (Pi) instead of water

Limit dextrin
what is left when phosphorylase can go no further (stubs of 4 at each branch)

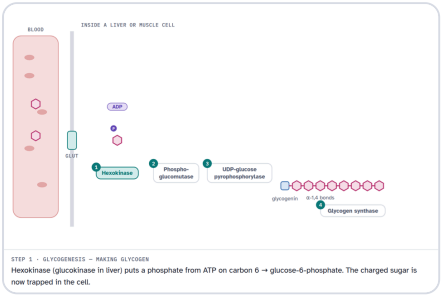
Allosteric regulation
switching an enzyme by a small molecule binding somewhere other than the active site

Futile cycle
two opposite pathways running at once, wasting energy

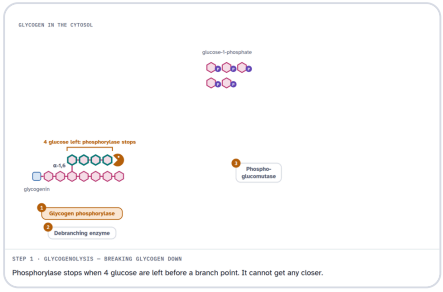
WATCH

The animations

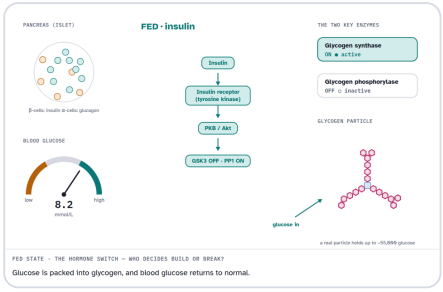
A PDF can't move, so the animations live next to it. Scan the code, or open `glycogen-animations.html`, `glycogen-animations.mp4` or the three GIFs in this folder. Each scene has captions and you can jump to any step.



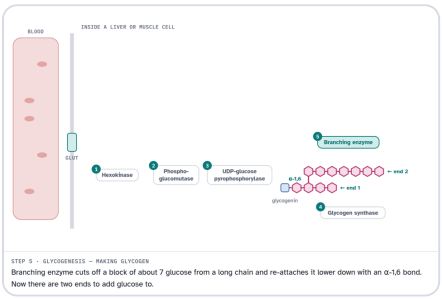
1 · Build: hexokinase adds a phosphate from ATP.



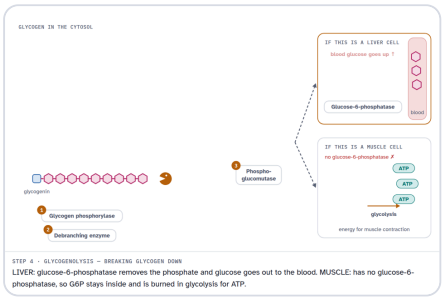
2 · Break: phosphorylase stops 4 before the branch.



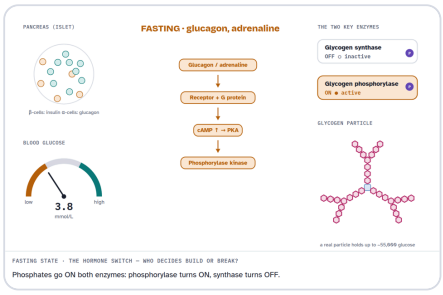
3 · Fed: insulin → PP1 removes phosphates; synthase ON.



1 · Branching enzyme makes an α-1,6 branch.



2 · Liver sends glucose to blood; muscle burns G6P.



3 · Fasting: glucagon → cAMP → PKA; phosphorylase ON.

Scan to play the animations

`moel80.github.io/jana-study-hub/study-packs/biol214/glycogen/glycogen-animations.html`

PRACTICE

Test yourself

1. Which enzyme is the controlled (rate-limiting) step of glycogenesis?
   a) glycogen phosphorylase
   b) glycogen synthase
   c) branching enzyme
   d) phosphoglucomutase

2. Glycogen phosphorylase breaks α-1,4 bonds using:
   a) water
   b) ATP
   c) inorganic phosphate (Pi)
   d) UTP

3. Glucagon stimulates glycogenolysis in:
   a) liver only
   b) muscle only
   c) both liver and muscle
   d) neither

4. Which statement is TRUE?
   a) phosphorylated glycogen synthase is active
   b) phosphorylated glycogen phosphorylase is active
   c) insulin adds phosphates to both enzymes
   d) PKA activates glycogen synthase

5. Which vitamin-derived helper does glycogen phosphorylase need?
   a) NAD^+ (niacin)
   b) FAD (riboflavin)
   c) PLP (vitamin B6)
   d) biotin

6. What is the activated glucose donor for glycogen synthase, and what happens to the PPi made alongside it?

7. Why can muscle glycogen NOT be used to raise blood glucose?

8. How many glucose are left on a branch when phosphorylase stops, and what does debranching enzyme do with them?

9. Why does glycogen have so many branches? Give two reasons.

10. A 20-year-old gets painful cramps and dark urine after sprinting. Blood glucose is normal, and blood lactate does not rise during an exercise test. Which enzyme is missing, and why doesn't lactate rise?

11. A baby has a very large liver, low blood glucose after a few hours without feeding, and lactic acidosis. Name the disease and the missing enzyme.

12. Compare the hormonal control of liver and muscle glycogen. (6 marks)

PRACTICE

Answers

1. **b)** Glycogen synthase.
2. **c)** Inorganic phosphate. This is phosphorolysis, and it gives glucose-1-phosphate.
3. **a)** Liver only. Muscle has no glucagon receptors; adrenaline acts on both.
4. **b)** Phosphorylated phosphorylase (the “a” form) is active. Phosphorylated synthase is inactive.
5. **c)** Pyridoxal phosphate (PLP), made from vitamin B6.
6. UDP-glucose (from glucose-1-P + UTP, by UDP-glucose pyrophosphorylase). PPi is split into 2 Pi by pyrophosphatase, which makes the reaction effectively irreversible.
7. Muscle lacks glucose-6-phosphatase, so it cannot turn G6P into free glucose, and G6P cannot cross the cell membrane. Muscle uses its glycogen for its own glycolysis.
8. 4 glucose. The transferase activity moves 3 to the end of another chain; the α-1,6-glucosidase activity releases the last one as free glucose.
9. (1) Many non-reducing ends, so many enzymes can add or remove glucose at once: faster storage and release. (2) Higher solubility. (Also: packs lots of glucose into few particles, so little osmotic effect.)
10. Muscle glycogen phosphorylase: McArdle disease (GSD V). Muscle can't break down its glycogen, so there is no glucose-6-P to feed anaerobic glycolysis and no lactate is made. Liver phosphorylase is normal, so blood glucose is fine.
11. Von Gierke disease (GSD type I): glucose-6-phosphatase deficiency. The liver can make G6P but can't release glucose, so fasting hypoglycaemia occurs; the extra G6P is pushed into glycolysis → lactate.
12. Both: insulin (fed) activates PP1 and inactivates GSK3 via PKB, dephosphorylating synthase (ON) and phosphorylase (OFF). **Liver:** glucagon (fasting) and adrenaline (stress) act via GPCR → cAMP → PKA → phosphorylase kinase → phosphorylase a; glucose itself inhibits liver phosphorylase a (glucose sensor); the liver releases free glucose into blood. **Muscle:** no glucagon receptors, so it responds to adrenaline (same cAMP cascade) plus Ca^{2+} (contraction) and AMP (low energy); it keeps the glucose for its own glycolysis.

How did you go?

Mark any you got wrong, then re-read that page and watch the matching animation. The Study Hub flashcards and practice quiz for BIOL214 cover the same topic (“Gluconeogenesis and glycogen metabolism”).

SUMMARY

One-page summary

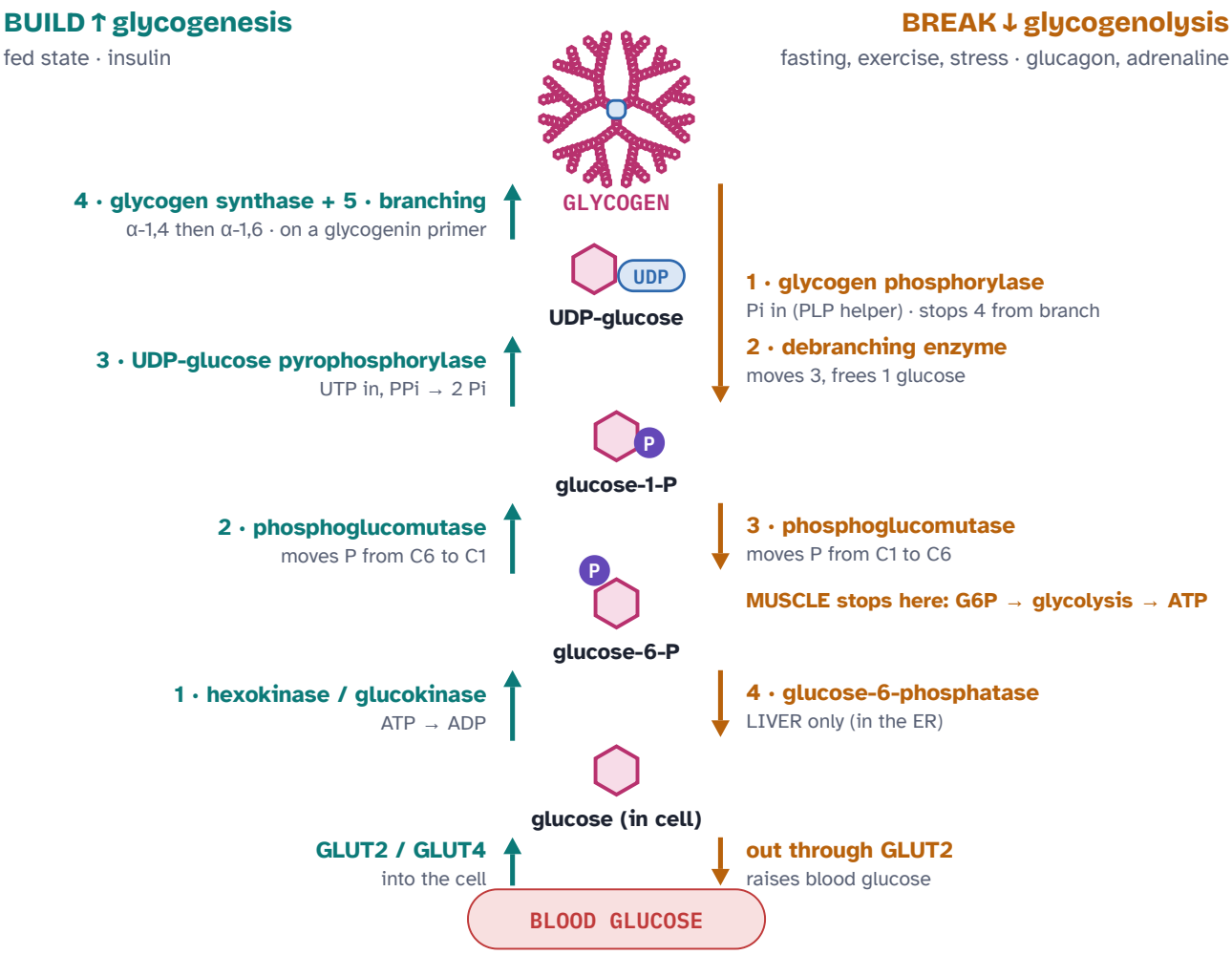


Fig. 17 · The whole topic in one picture: build on the left, break on the right.

Structure

Branched glucose polymer: α-1,4 chains, α-1,6 branches every 8–12. Built on glycogenin. Liver ≈ 100 g, muscle ≈ 400 g.

Control

Insulin → phosphate OFF → synthase ON. Glucagon/adrenaline → cAMP → PKA → phosphate ON → phosphorylase ON. Muscle: + Ca^{2+}, + AMP. Liver: glucose inhibits phosphorylase.

Clinical

von Gierke (G6Pase): fasting hypoglycaemia, big liver. McArdle (muscle phosphorylase): exercise cramps, no lactate rise. Pompe (lysosomal): heart and muscle.

CREDITS

Photo credits and sources

Photos are used under the licences shown. Photos were resized; nothing else was changed. All diagrams and animations were drawn for this pack.

Image	Author / source	Licence	Where to find it
Liver biopsy, PAS stain; PAS after diastase	Department of Pathology, Calicut Medical College	CC BY-SA 4.0	`Wikimedia Commons: File:Glycogen_storage_disorder_(PAS_stain).jpg and …_(PAS_with_diastase).jpg`
Glycogen space-filling model	CeresVesta (English Wikipedia)	Public domain	`Wikimedia Commons: File:Glycogen_spacefilling_model.jpg`
Glycogenin structure (PDB 1LL2)	RCSB Protein Data Bank; Gibbons, Roach & Hurley (2002) J Mol Biol	CC0 (RCSB PDB images)	`rcsb.org/structure/1LL2`
Glycogen phosphorylase b structure (PDB 1GPB)	RCSB Protein Data Bank; Acharya et al. (1991)	CC0 (RCSB PDB images)	`rcsb.org/structure/1GPB`
Spaghetti Bolognese	Wikimedia Commons user Chingon	Public domain	`Wikimedia Commons: File:Spaghetti_Bolognese.jpg`
100 m heat, Moscow 2013	Tobi 87	CC BY-SA 3.0	`Wikimedia Commons: File:Leichtathletik_WM_2013_Moskau_100_m_Vorlauf.jpg`
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Glucagon emergency rescue kits	Mizi m	CC0	`Wikimedia Commons: File:Glucagon_emergency_rescue_kits_image.jpg`
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Cori disease liver, H&E	Nephron	CC BY-SA 3.0	`Wikimedia Commons: File:Glycogen_storage_disease_in_liver_-_high_mag.jpg`
Pompe disease muscle biopsy	Jensflorian	CC BY-SA 3.0	`Wikimedia Commons: File:Pompe_vacuoles.jpg`
Myozyme	Genzyme	Public domain	`Wikimedia Commons: File:Myozyme.jpg`

Where the facts come from

- BIOL214 subject outline and lecture topics, Spring 2026 (Week 4: gluconeogenesis and glycogen metabolism).
- Nelson & Cox, *Lehninger Principles of Biochemistry*, chapters on carbohydrate metabolism and its regulation.
- Berg, Tymoczko, Gatto & Stryer, *Biochemistry*, chapter on glycogen metabolism.
- Sports-nutrition numbers (carb loading, running energy cost) are typical textbook values and are rounded.

Made for Jana

Part of the Jana Study Hub. If something here doesn't match your lecture slides, **trust the slides** for the exam, and flag it so this pack can be fixed.